NARRATIVE SYNTHESIS · INTRAVENOUS LIDOCAINE IN ADULT BURNS

Lidocaine infusions in adult burn patients: analgesic efficacy, opioid sparing, and the *unresolved* question of long-term outcomes.

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01 · BACKGROUND & PURPOSE

Why revisit lidocaine for burn pain?

Pain after burn injury is **severe, prolonged and frequently under-controlled** by opioid-based regimens alone, with consequences extending well beyond the acute admission — opioid-related adverse effects, tolerance, chronic pain, and impaired functional recovery.[9,13,19,20]

Intravenous lidocaine infusions have established **analgesic and opioid-sparing effects** in perioperative and non-burn contexts.[15,16,18] Their use in burn care is expanding, but the evidence base supporting that uptake remains **small, short, and inconsistently positive.**[7,8]

Purpose. Synthesise the current clinical evidence for IV lidocaine in adult burn patients, and identify gaps relating to pain control, opioid use, safety, and patient-centred long-term outcomes.

AT A GLANCE

What this synthesis covers

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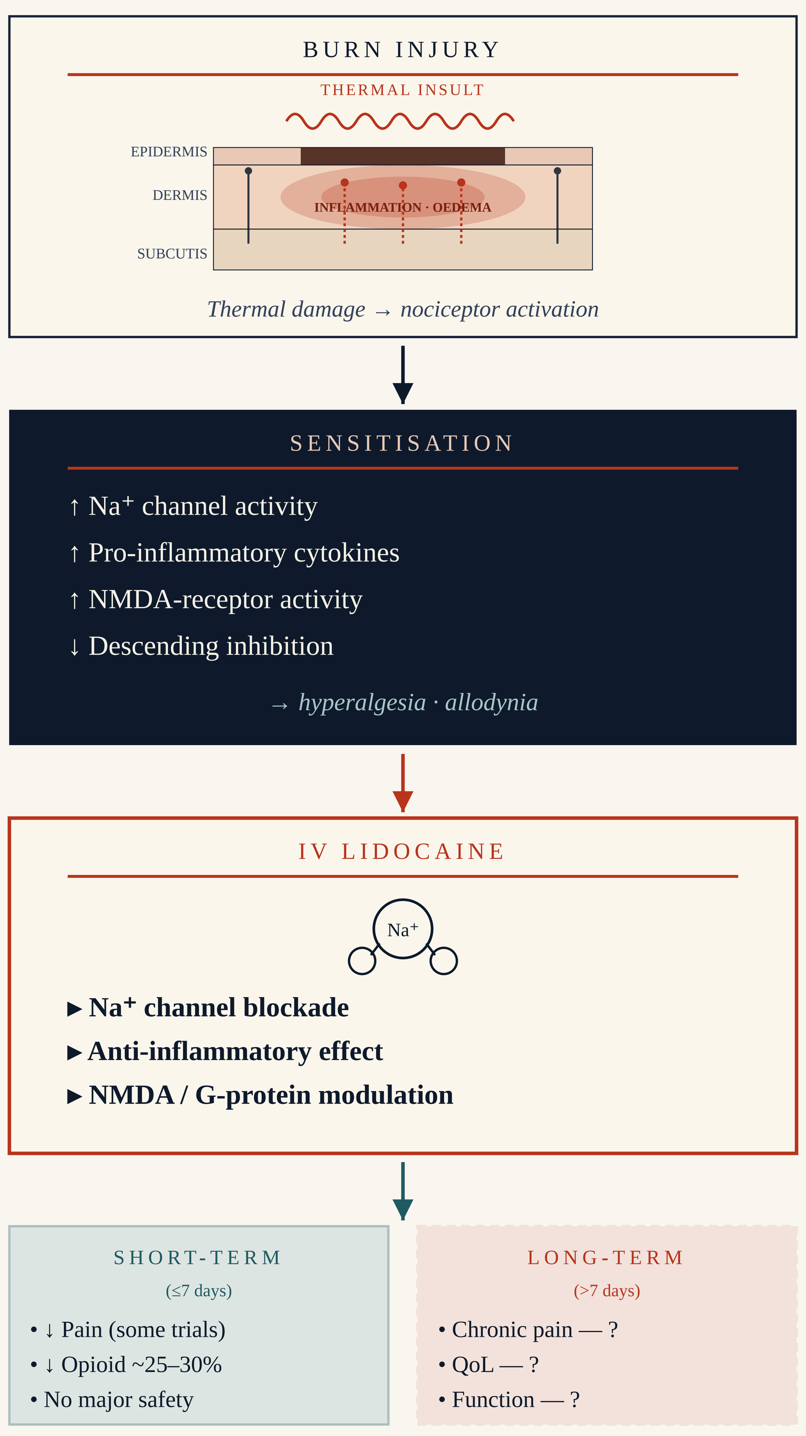
ELIGIBLE STUDIES	RANDOMISED TRIALS	MAXIMUM FOLLOW-UP	QOL OR FUNCTIONAL DATA

Adult burn patients only. Outcomes assessed: **pain intensity**, **opioid consumption**, adverse effects, quality of life, functional recovery.

02 · MECHANISM

Why lidocaine might work — and where the argument breaks down



Plausible mechanism — inconsistent short-term signal — absent long-term data.

03 · METHODS

Narrative synthesis

01 **Design.** Narrative synthesis. RCTs, observational studies, and SRs/meta-analyses eligible.

02 **Population.** Adult burn patients (≥16y) receiving IV lidocaine infusion.

03 **Intervention.** IV lidocaine vs standard analgesia, placebo, or active comparator.

04 **Outcomes.** Pain, opioid use, adverse effects, QoL, functional recovery.

05 **Synthesis.** Findings grouped by outcome domain; consistency, magnitude, and gaps appraised narratively.

04 · INCLUDED STUDIES

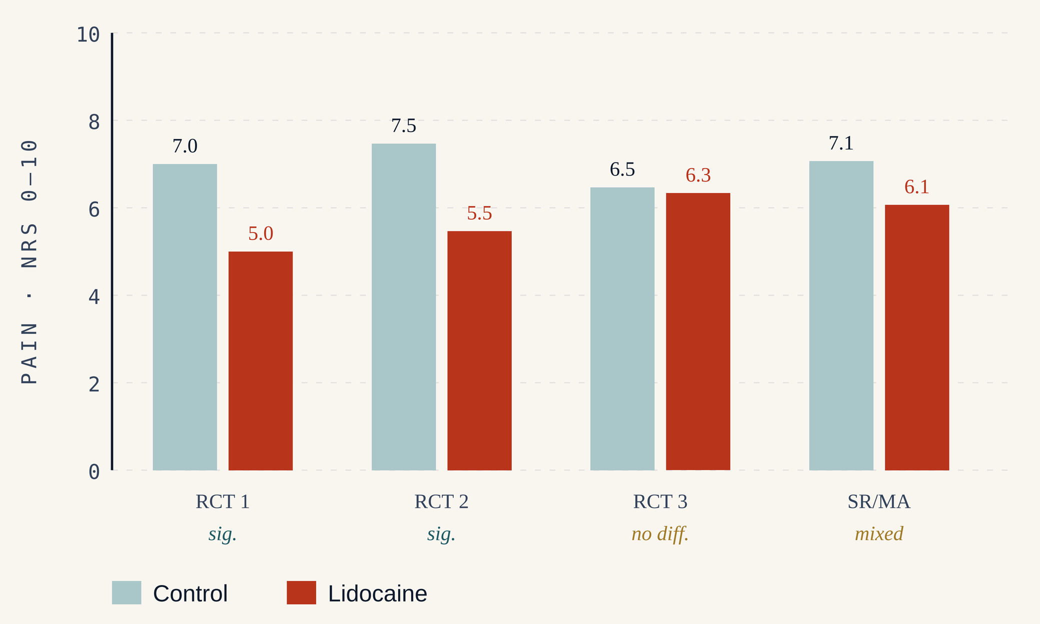
Small samples, short horizons, mixed signals

STUDY	SETTING	PAIN	OPIOID
JÖNSSON 1991[1]	Case series n=7	↓ MARKED	OPIATE-FREE
CASSUTO 2003[2]	Case series	↓ MARKED	OPIATE-FREE
WASIAK 2011[4]	RCT n=45 procedural	↓ SIG.	NO DIFF.
ABDELRAHMAN '20[5]	RCT n=19, 7-day	NO DIFF.	↓ ~25%
HAGHIGHI 2023[6]	RCT n=66 acute	↓ SIG.	↓ SIG.
COCHRANE SR[7]	Wasiak 2014	INSUFF.	INSUFF.
META-ANALYSIS[8]	Elgadi 2025 (4 RCTs, n=152)	NO SIG.	SMALL, NS
NARRATIVE[13]	Morgan 2018	*Calls for higher-quality evidence.*	

*Follow-up: procedural timepoints to **7 days** max. None reported QoL or functional recovery.[7,8]*

05 · PAIN INTENSITY

Real but inconsistent signal



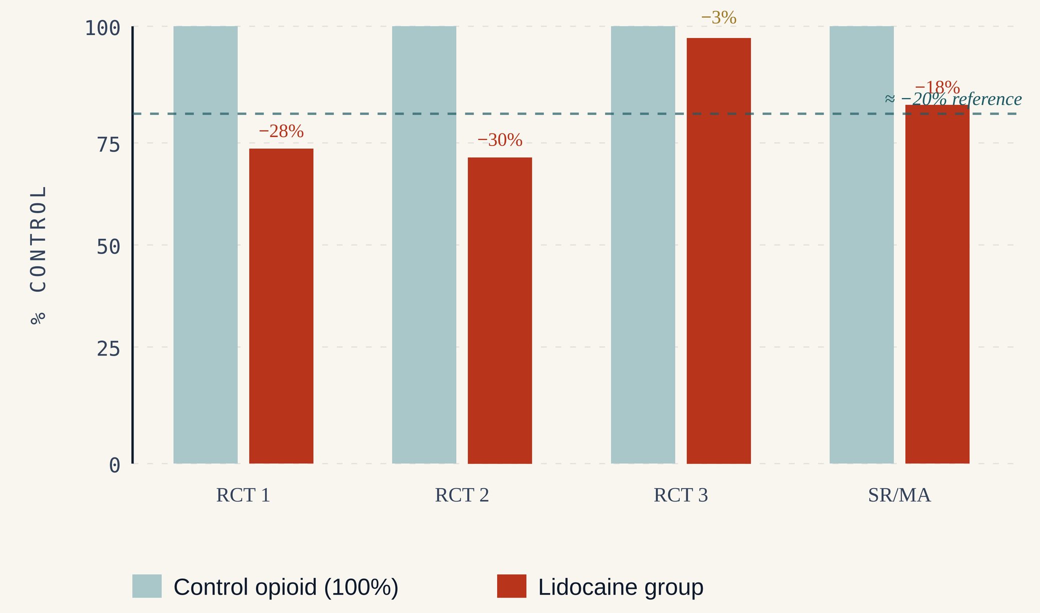
Reported NRS pain scores; magnitudes vary by setting and infusion regimen.

EVIDENCE GAP, DECLARED UPFRONT

No included study reported quality-of-life or functional recovery outcomes; follow-up never exceeded the acute inpatient phase.

06 · OPIOID SPARING

~25–30% reduction — when present



Where present, opioid-sparing effect clusters around 25–30%; in negative trials, effect is essentially absent.

07 · SAFETY

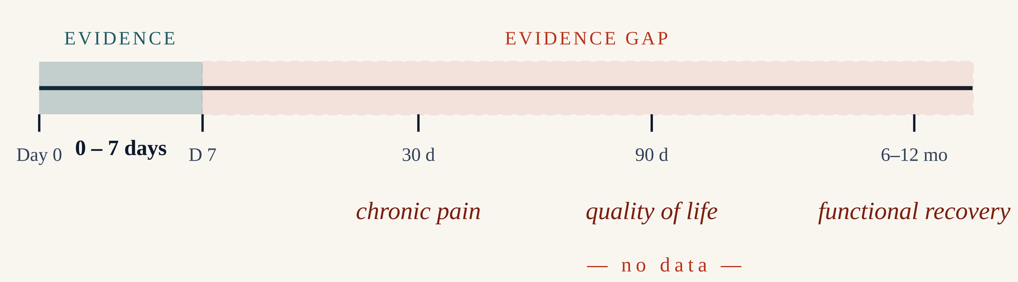
No new signal — but underpowered

Across included studies, no significant safety concerns were attributable to IV lidocaine infusion.[4,5,6,8] Plasma concentrations remained within accepted therapeutic ranges; serious adverse events not increased over standard care.

However: small samples, brief follow-up, non-standardised protocols. The literature is **underpowered to detect rare adverse effects** — cardiovascular toxicity, neurotoxicity, drug interactions.[17] *Reassuring, not definitive.*

08 · THE FOLLOW-UP GAP

Where the evidence ends — and patients still live



Every included study stops within the first week. The outcomes that matter most to burn survivors — **persistent pain, QoL, return to function** — sit entirely outside the current evidence base.[19,20]

09 · CONCLUSION & IMPLICATIONS

A short-term signal, a long-term silence.

> *"Intravenous lidocaine infusions may offer real short-term analgesic and opioid-sparing benefits in selected adult burn patients — but the question of whether they change how those patients **live, recover, and return to function** remains unanswered."*

FOR THE CLINICIAN

IV lidocaine is reasonable as an opioid-sparing adjunct in selected adult burn patients, within established safety monitoring. It should not be assumed to alter long-term outcomes.

FOR RESEARCH

Future trials must be adequately powered, prospectively designed, and incorporate **extended follow-up** (≥3 months) with patient-centred outcomes.

FOR THE BBA COMMUNITY

A coordinated UK multi-centre approach across specialist burn services could rapidly close the long-term evidence gap that limits current recommendations.

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